DAY 02 ASSIGNMENT
Understanding my network

Submitted by

SREEKAKULAM THULASIRAM

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Topic: Computer Networking Basics and Network Setup

1) THEORY

1) What is a computer network?

A computer network is when two or more computers or devices are connected to exchange information, resources and services via either a wired or wireless connection. An example would be multiple computers connected to the same Wi-Fi.

2) What is the difference between client and server?

A client is a piece of software/device that requests information or services. A server is a computer/system that accepts requests from clients and returns the requested information or service. For example, browsers are clients and web servers provide the information from websites.

3) What is a LAN?

LAN stands for Local Area Network. It connects computers and devices together in a local area, such as a home, office, lab or college. This allows devices to connect and share resources.

4) What is a WAN?

Wide Area Network. This connects multiple computers and networks over a large area. The internet is the most common example of a Wide Area Network.

5) What does a switch do?

A switch connects multiple devices within the local area network. It uses MAC addresses to direct frames to the correct device.

6) What does a router do?

A router connects different networks. It routes data packets between networks. An example of a router in a home would be the router connecting devices in the local network to the internet. Routers use IP addresses to tell where to send packets.

7) What is a firewall?

A firewall is a security system that filters traffic coming in and out of a network based on set security rules. This will help stop unauthorised access to your computers/network.

8) What is an IP address?

An IP address is a numerical address given to a network interface on a device. This helps identify the device on an IP network. There are two types of IP addressing, IPv4 and IPv6.

9) What is a MAC address?

A MAC (Media Access Control) address is an address given to a network interface. They are mostly used to identify a device. Ethernet MAC addresses are often 48 bits.

10) What is a default gateway?

This is the device (usually a router) that your computer will use to send network traffic. This is used when trying to connect to something not on your local network.

11) What is DNS?

Domain Name System. This translates domain names (e.g. google.com) into IP addresses so computers can communicate with each other. This makes it easier for people to access websites.

12) Why is networking knowledge important for cyber security?

Understanding networking is important for cyber security as most cyber attacks involve some form of communication between devices and networks. Knowing about IP addresses, ports, protocols, routers, DNS and network traffic can help security analysts identify any suspicious connections, identify unauthorised access and investigate any cyber attacks. This can also help with setting up firewalls.

2) MY NETWORK INFO

I used the windows command prompt to view my network settings by using the command 'ipconfig /all'.

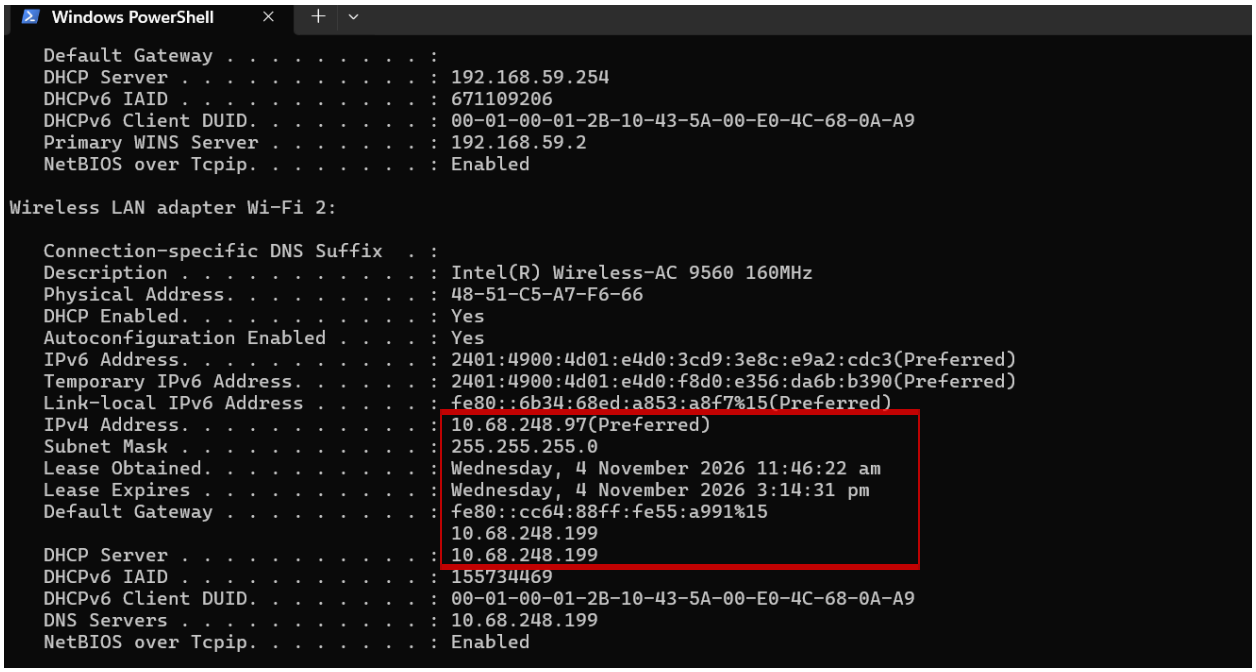
The below information was taken from my active network adapter:

Network Info	**My Computer**
Private IPv4 Address	10.68.248.97
Default Gateway	10.68.248.199
DNS Server	10.68.248.199

3) SCREENSHOTS

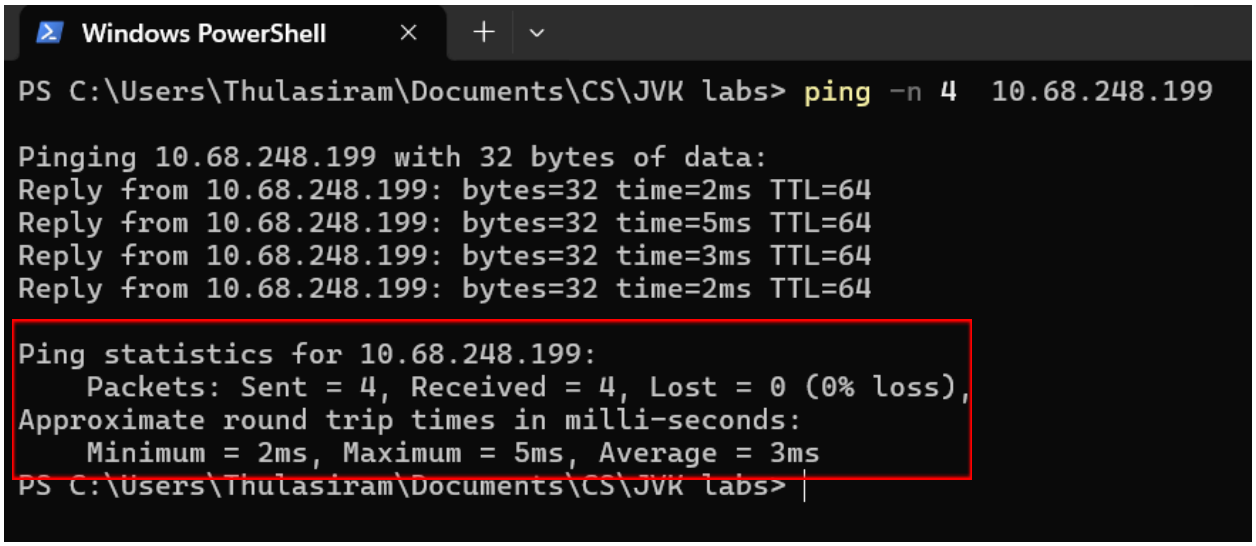
Screenshot 1 – IP Config

The below screenshot shows the network information from running the 'ipconfig /all' command. This shows my private IPv4 Address, default gateway and DNS Server.



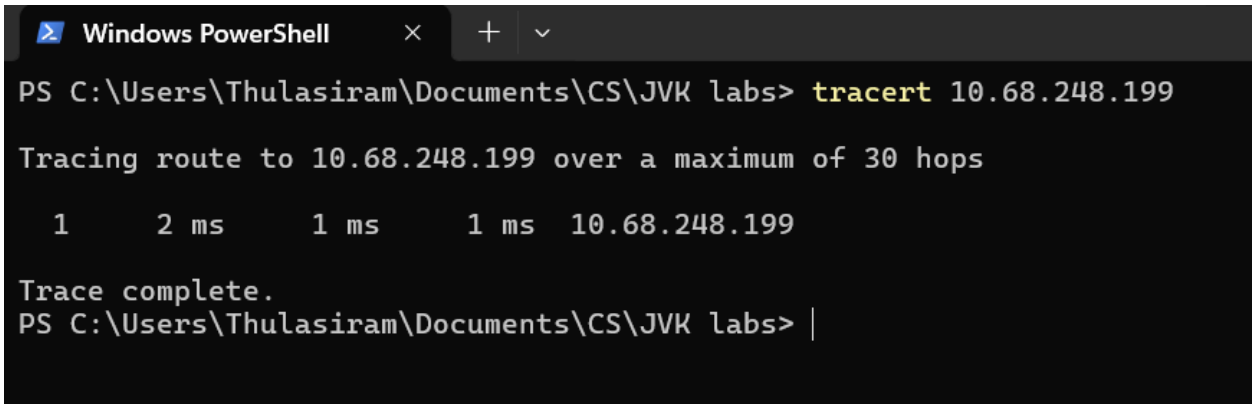
Screenshot 2 – Ping Command

The below screenshot shows the results from running the ping command on my default gateway. This can show if my computer can communicate with my local router.



Screenshot 3 – Traceroute Command

The below screenshot shows the results from running the tracert command on my default gateway. This shows the hops between my computer and local router.



CONCLUSION

By completing this assignment I have learnt the basics of computer networking. I now know what a LAN, WAN, switch, router, firewall, IP address, MAC address and DNS are. I have also learnt how to view my network settings using the command prompt and how to use the ping and tracert commands. This knowledge can help me understand how networks work and how cyber security professionals can monitor and secure them.